

Dr. Wenyue Zou

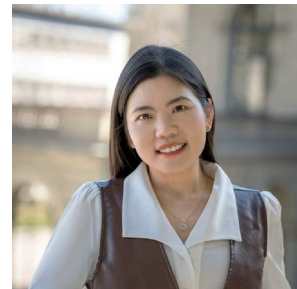
Ph.D. in Environmental Science

Postdoc, Swiss Federal Institute of Aquatic Science and Technology

[Expertise Center for Climate Extremes](#), University of Lausanne

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RESEARCH PROFILE

I am an environmental scientist working on extreme precipitation analysis, stochastic rainfall generation, and flood-risk assessment under climate change. My research focuses on generating high-resolution rainfall fields and translating rainfall uncertainty into flood-hazard and risk-relevant information. I am particularly interested in blue-green infrastructure design for climate-resilient urban water management.

PROFESSIONAL SKILLS

- Extreme precipitation and climate extremes analysis
- Observation-based projection of future high-resolution rainfall fields under climate warming
- Geostatistical spatial downscaling and stochastic storm modelling
- Climate-informed generation of future rainfall scenarios and storm-flood risk assessment
- Scientific programming: Python, MATLAB, R, ArcGIS/QGIS
- Climate data science: station, radar, satellite, reanalysis, climate-model, and hydrological datasets

TRAINING AND EMPLOYMENT HISTORY

2026.06 – present **Postdoctoral researcher**, Swiss Federal Institute of Aquatic Science and Technology

2024.08 – 2026.08 **Visiting Researcher**, Institute of Environmental Engineering, ETH Zurich

Developing observation-based future high-resolution storm projection for flood-risk assessment

2022 – 2026.06 **Doctoral Researcher**, Faculty of Geosciences and Environment, University of Lausanne

2022 – 2023 **Teaching Assistant**, Watershed and River Network Modelling, University of Lausanne

2019 – 2021 **Research Assistant**, Faculty of Geographic Science, Beijing Normal University

2019 – 2020 **Teaching Assistant**, Assessment of Climate Change and its Impacts, Beijing Normal University

EDUCATION

2022 – 2026

Ph.D. in Environmental Science

Expertise Center for Climate Extremes (ECCE)

Faculty of Geosciences and Environment, University of Lausanne, Switzerland

Thesis: Future Changes in Rainfall Properties and Their Effects on Urban Flooding

Supervisor: Nadav Peleg

2018 – 2021

M.Sc. in Physical Geography

Faculty of Geographic Science, Beijing Normal University, China

GPA: 3.75/4.0 (top 10%)

Thesis: Event-Scale Spatiotemporal Characteristics of Rainfall Based on a High-Density Rain Gauge Network

Supervisor: Shuiqing Yin

2014 – 2018

B.Sc. in Geography

Faculty of Geographic Science, Northwest Normal University, China

GPA: 3.84/4.0 (top 1%)

Thesis: Spatiotemporal Variability of Hourly Precipitation in the Haihe River Basin over 1961–2012

Supervisor: Junju Zhou

SELECTED PUBLICATIONS

First-author publications:

- [4] **Zou, W., et al.** 2025. A process-based framework linking temperature-rainfall projections and urban flood modeling. *Hydrology and Earth System Sciences*, under review. <https://doi.org/10.5194/egusphere-2025-4099>.
- [3] **Zou, W.,** Wright, D. B., Peleg, N. 2025. Morphing sub-daily rainfall fields based on temperature shifts to project future changes in rainfall extremes. *Water Resources Research*, <https://doi.org/10.1029/2024WR038847>.
- [2] **Zou, W., et al.** 2024. Multiple-point geostatistics-based spatial downscaling of heavy rainfall fields. *Journal of Hydrology*. <https://doi.org/10.1016/j.jhydrol.2024.130899>.
- [1] **Zou, W., et al.** 2021. Spatial interpolation of the extreme hourly precipitation at different return levels in the Haihe River basin. *Journal of Hydrology* 598, 126273. <https://doi.org/10.1016/j.jhydrol.2021.126273>.

Co-author publications:

- [8] Nakhael, N., Li, R., Chen, Y., **Zou, W.**, Ni, G., 2025. **Scalable Urban Pluvial Flood Prediction with Tile-Based Super-Resolution Modeling**. *Geoscience Frontiers*, under review.
- [7] Zhu, Y., Burlando, P., Zhang, Y., Wang, J., **Zou, W.**, Fatichi, S., 2025. Influence of Urban Landscape Morphology Changes During Urban Expansion on Pluvial Flooding. *Sustainable Cities and Society*. <https://doi.org/10.1016/j.scs.2025.107018>
- [6] Li, T., Yin, S., Xiao, Y., Wang, M., **Zou, W.**, Peleg, N., 2025. Quantifying rainfall variability and potential hazards of extreme events in Beijing through stochastic simulations. *Journal of Hydrology: Regional Studies*. <https://doi.org/10.1016/j.ejrh.2025.103068>
- [5] Li, Q., Zhou, J., **Zou, W.**, et al., 2020. A tributary-comparison method to quantify the human influence on hydrological drought. *Journal of Hydrology*. <https://doi.org/10.1016/j.jhydrol.2020.125652>.

THREE MAJOR RESEARCH ACHIEVEMENTS

Achievement 1. Generating long-term high-resolution spatial rainfall fields for urban hydrological applications

A key challenge in urban hydrology and wastewater catchment management is limited availability of long-term, high-resolution spatial rainfall data. Sparse rain-gauges and coarse satellite products often cannot fully capture realistic storm structures that are important for flood risk assessment, combined sewer overflow analysis, and blue-green infrastructure design^[1, 2]. During my MSc and PhD, I developed statistical and geostatistical methods to improve the spatial representation of rare rainfall extremes from limited observations and remotely sensed datasets^[1, 2].

I first evaluated different **spatial interpolation methods** using hourly extreme precipitation from 230 stations and showed that Kriging with external drift can improve return-level mapping across a large river basin^[1]. I then developed a **multiple-point geostatistics-based downscaling framework** to reconstruct realistic high-resolution heavy-rainfall fields from coarse satellite products using radar-derived spatial patterns as training data. Applied to Beijing, this framework downscaled 22 years of hourly satellite rainfall from 8 km to 1 km while preserving both rainfall intensity and spatial storm structure^[2].

Together, **these methods provide a strong basis for developing representative rainfall products for urban catchments**. They are directly relevant to gauge-network design, radar-informed rainfall reconstruction, and the generation of spatially realistic rainfall inputs for urban hydrological and wastewater applications^[1-2, 4, 8].

Achievement 2. Projecting future high-resolution, sub-daily rainfall fields under climate warming

Projecting future precipitation fields requires not only estimating changes in rainfall intensity, but also capturing how storm extent, spatial variability, and event rarity may shift under climate warming^[4]. This is particularly important for urban wastewater systems, where short-duration extremes and spatial rainfall structure can strongly influence CSO occurrence and performance. **However, future sub-daily rainfall projections remain limited** because **convection-permitting climate** models are computationally demanding and often provide relatively short simulations.

During my PhD, I **developed a process-informed framework** to project future high-resolution rainfall fields under different warming scenarios. Specifically, I developed a distribution-based spatial quantile mapping method that **uses rainfall-temperature scaling relationships to perturb observed rainfall fields**. By integrating this method with stochastic storm transposition (SST), I generated ensembles of physically plausible future storm realizations and quantified spatial changes in hourly rainfall extremes under multiple regional warming levels^[3].

This work produced a data-driven, physically grounded, and scalable framework for generating future rainfall scenarios. It supports uncertainty-aware hazard assessment by representing changes in rainfall intensity, storm spatial patterns, and storm-location uncertainty. It also provides a methodological foundation for developing CPM-informed DSQM rainfall projections and long synthetic rainfall ensembles for climate-resilient urban wastewater planning^[3-4, 7-8].

Achievement 3. Linking extreme-precipitation science with impact-relevant flood-risk assessment

A further challenge in climate-risk research is translating projected changes in precipitation into impact-relevant information while preserving uncertainty from the rainfall generation stage. During my PhD, I applied future storm scenarios to high-resolution hydrodynamic flood modelling to examine **how changes in storm intensity and spatial patterns can influence flood depth, inundation extent, and spatial exposure under climate warming**^[4].

This work demonstrated how uncertainty in future rainfall extremes can be propagated into impact-relevant flood-hazard and risk assessment. Although my previous applications focused mainly on urban pluvial flooding, the same rainfall-to-impact framework can be extended to wastewater catchment management, including CSO-relevant rainfall indicators, stress-testing of drainage systems, and evaluation of climate-adaptation options.

Beyond individual publications, I have also contributed to **interdisciplinary climate-risk research** through the Expertise Center for Climate Extremes (ECCE, UNIL), the UNIL-EPFL Center for Climate Impact and Action, peer review, and collaborative projects on climate extremes, infrastructure impacts, and stakeholder-relevant risk assessment. These experiences have strengthened my ability to **connect rigorous climate-extremes research with practical questions of uncertainty, adaptation, and decision-relevant climate information**.

SERVICES

- Peer review: *Earth System Science Data*, *Journal of Hydrology*, *Scientific Reports*, *Natural Hazards*
- 2024-Present PhD Committee Member, Expertise Center for Climate Extremes, University of Lausanne
- 2023 Student Committee Member, CLIMACT, UNIL-EPFL
- 2021 Student Committee Member, Association du Corps Intermédiaire, FGSE, UNIL

GRANTS, AWARDS, AND SCHOLARSHIPS

- 2026 Innosuisse Innovation Cheque Project (**15k CHF**)
- 2026 Doctoral Scholarship, Société Académique Vaudoise (**6000 CHF**)
- 2025 Swiss-China Innovation & Entrepreneurship International Competition, Global Finalist (*top 15%*)
- 2021 Excellent Master's Thesis Award, Beijing Normal University
- 2018, 2019 Academic Scholarship at Beijing Normal University (*First Class, twice, top 10%*)
- 2017 First Prize in Scientific Research Challenge Cup at Northwest Normal University (*top 1%*)
- 2016 First Prize in National College Students Mathematical Modeling Competition, China (*top 10%*)
- 2015 National Endeavor Scholarship, China (*top 10%*)